
Climate Risk & Financial Tone Analytics

Process Documentation

Data Sources, Cleaning & Integration Pipeline, and Dashboard Architecture for
Quarterly Earnings-Call Transcript Analysis of BSE-Listed Companies

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This document describes, end to end, the two source datasets that feed this project, the pipeline that cleans and merges them into a single canonical dataset, and the dashboard built on top of that dataset to explore climate-risk exposure and financial tone across quarterly earnings calls of Bombay Stock Exchange (BSE) listed companies. It is intended as the reference record of the project's data and design decisions.

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1. Introduction & Purpose

Earnings-call transcripts are a rich, unstructured record of how a company's management communicates with investors each quarter — what they emphasize, how confidently they speak, and how they frame climate-related topics such as emissions, regulation, and physical or transition risk. This project analyzes quarterly earnings-call transcripts for approximately 1,363 BSE-listed companies, spanning 2011–2025, along two independent analytical dimensions:

- **Climate-risk exposure and framing** — how much of each call discusses climate topics, and whether that discussion is framed as risk, opportunity, or neutral fact.
- **General financial tone, hedging, and readability** — how positive or negative the language is overall, how much uncertainty/hedging is present, and how complex the language is.

Each dimension was produced by an independent scoring pipeline, resulting in two separate output datasets. This document explains both datasets in full column-by-column detail, the pipeline that merges and cleans them into one canonical dataset, and the dashboard built to let a user explore the result — including exactly how each underlying column surfaces in the dashboard's features.

2. Dataset One: Climate Exposure & Sentiment Scores

2.1 Overview & Methodology

This dataset scores each transcript for climate relevance and climate sentiment using a transformer-based language model purpose-built for climate-related text (further pretrained on a large corpus of climate news, scientific abstracts, and corporate sustainability reports, on top of a general-purpose language model). Two classifiers are applied to every transcript:

- **Climate-relevance classifier** — applied at the sentence level, labels each sentence as climate-related or not.
- **Sentiment classifier** — applied only to sentences already labeled climate-related, further classifies each as framing a *risk*, an *opportunity*, or a *neutral* statement.

The raw output file, **ClimateBert_Scores_Output_Updated.csv**, contains 15,812 rows across 12 columns — one row per transcript file — covering 1,362 distinct companies (identified by BSE security code).

2.2 Column-by-Column Reference

Column	Type	Description
Filename	text	Source transcript file name. Pattern: SecurityCode_Year_Quarter, with an optional trailing part-number suffix (e.g. 534139_2025_Q4.txt). This is the join key to Dataset Two.
Security_Code	integer	BSE security / scrip code uniquely identifying the company.
Company_Name	text	Full legal name of the company.
NSE_Name	text	Short ticker-style name as listed on the National Stock Exchange.
Year	integer	Fiscal year of the earnings call, parsed from the filename.

Column	Type	Description
Quarter	text	Fiscal quarter: Q1, Q2, Q3, or Q4. Two further values appear due to filename-parsing limitations: 'Unknown' (no recognizable year/quarter pattern in the filename) and 'Q#.error' (a partially recognized but malformed pattern).
Total_Sentences	integer	Total number of sentences in the transcript after sentence-level tokenization.
Climate_Sentences_Count	integer	Number of sentences classified as climate-related by the relevance classifier.
Climate_Exposure_Score	decimal	$\text{Climate_Sentences_Count} \div \text{Total_Sentences} \times 100$. The percentage of the call devoted to climate topics. Verified to hold exactly across every row of this dataset.
Sentiment_Risk_Count	integer	Number of climate sentences framed as a risk (e.g. transition risk, physical risk, regulatory exposure).
Sentiment_Opportunity_Count	integer	Number of climate sentences framed as an opportunity (e.g. new markets, efficiency gains, green financing).
Sentiment_Neutral_Count	integer	Number of climate sentences that are purely factual/descriptive, with no risk or opportunity framing.

2.3 Data Characteristics

Quarter distribution — Q1: 4,332, Q2: 4,080, Q3: 3,705, Q4: 3,666, Unknown: 20, plus 9 rows carrying a '.error' suffix on the quarter value. Two rows carry an implausible Year value (year numbers outside any real fiscal range), both on rows where Quarter is 'Unknown' — both are filename-parsing artifacts, addressed in the cleaning pipeline (Section 5).

Two derived-column invariants were verified exactly against every row in this dataset: Climate_Exposure_Score always equals $\text{Climate_Sentences_Count} \div \text{Total_Sentences} \times 100$, and Sentiment_Risk_Count + Sentiment_Opportunity_Count + Sentiment_Neutral_Count always equals exactly Climate_Sentences_Count. Both hold with zero exceptions, confirming the sentiment classifier is applied to, and only to, the sentences already flagged as climate-related.

3. Dataset Two: Financial Tone, Readability & FinBERT Scores

3.1 Overview & Methodology

This dataset scores the same transcripts for general financial sentiment, hedging language, and readability, using two complementary techniques rather than one:

- **Loughran-McDonald Master Dictionary** — a finance-specific word list (not a generic sentiment lexicon) used because ordinary words carry different connotations in financial disclosure than in everyday language (e.g. 'liability' reads as negative in a financial context but neutral elsewhere). Each transcript's words are counted against four categories: Positive, Negative, Uncertain, and Weak Modal (hedging words such as 'could', 'might', 'may').
- **FinBERT** — a pretrained finance-domain sentiment model, run independently on the same text to provide a model-based sentiment read for comparison against the dictionary-based one.

Readability is measured two ways: the Gunning Fog Index (a standard academic readability formula based on sentence length and syllable complexity) and a simple file-size-based verbosity proxy.

The raw output file, **merged_quarterly_transcripts_results_new.csv**, contains 15,828 rows across 20 columns — one row per transcript file — covering 1,363 distinct companies.

3.2 Column-by-Column Reference

Column	Type	Description
Company_ID	integer	BSE security code. Same numbering scheme as Security_Code in Dataset One.
Company_Name	text	Full legal name of the company.
Year / Quarter	integer / text	Fiscal period parsed from the filename. This dataset's parser is more permissive than Dataset One's: it recognizes both filename layouts (Year-then-Quarter and Quarter-then-Year) and produces 'Unknown' rather than an error value when neither pattern matches.
File_Name	text	Source transcript file name. Join key to Dataset One (Filename).
Positive_Count / Negative_Count	integer	Count of words matching the Loughran-McDonald positive / negative word lists.
Uncertain_Count	integer	Count of words matching the Loughran-McDonald uncertainty word list (e.g. 'appears', 'possibly', 'uncertain').
Weak_Modal_Count	integer	Count of hedging modal words (e.g. 'could', 'might', 'depending', 'should').
Tone_Score	decimal	$(\text{Positive_Count} - \text{Negative_Count}) \div (\text{Positive_Count} + \text{Negative_Count})$, a net sentiment score ranging from -1 (entirely negative) to +1 (entirely positive).
Percent_Uncertain / Percent_Weak_Modal	decimal	Respective word count $\div$ Total_Words $\times$ 100 — the share of the transcript that is hedging/uncertain language.
Total_Words	integer	Total word count of the transcript after cleaning (lowercased, numbers and punctuation stripped).
File_Size_MB	decimal	Raw size of the source transcript file, in megabytes.
File_Size_Readability	decimal	$-\log(\text{File_Size_MB} + 1)$, a verbosity proxy: larger, more verbose filings score lower (more negative).

Column	Type	Description
Fog_Index	decimal	Gunning Fog readability score: $0.4 \times (\text{average words per sentence} + \text{percent of complex words, defined as words of three or more syllables})$. Approximate interpretation: ~12 = high-school reading level, ~16 = college level, 18+ = graduate level.
FinBERT_Positive / FinBERT_Negative / FinBERT_Neutral	decimal	Class probabilities (summing to ~1.0) output by the FinBERT model for the transcript.
FinBERT_Tone	decimal	Net FinBERT sentiment. Empirically verified to equal $\text{FinBERT_Positive} - \text{FinBERT_Negative}$ (matches to rounding precision across the dataset).

3.3 Data Characteristics

Quarter distribution — Q1: 4,337, Q2: 4,088, Q3: 3,710, Q4: 3,673, Unknown: 20. No '.error' values occur in this dataset's Quarter column.

16 rows have a null FinBERT_Tone (and correspondingly null FinBERT_Positive/Negative/Neutral). These same rows also carry Total_Words = 0 and zero counts across every Loughran-McDonald category, consistent with a shared upstream text-extraction failure for those specific source files rather than a scoring error. All such rows are also absent from Dataset One entirely (see Section 4).

4. Relationship Between the Two Datasets

The two datasets describe the same underlying set of transcripts and are designed to be joined together. Two keys connect them:

- **Filename** ↔ **File_Name** (primary join key) — every one of the 15,812 rows in Dataset One has an exact matching row in Dataset Two. Dataset Two additionally contains 16 rows with no counterpart in Dataset One — these are precisely the 16 extraction-failure rows described above, for which no climate score could be produced at all.
- **Security_Code** ↔ **Company_ID** (company key) — the same BSE numbering scheme. 1,362 of the two datasets' company codes overlap; the remaining handful appear in only one dataset, generally reflecting a company with no successfully scored transcripts in the other pipeline.

The intended grain of both datasets is one row per company, per fiscal year, per quarter. In practice this grain is not always unique: a meaningful number of company-quarters are represented by more than one transcript file, distinguished by a numeric suffix on the filename (for example, two files ending in '_1' and '_3' for the same company and quarter). The cause of this duplication — whether these represent separate segments of the same call (such as prepared remarks versus a question-and-answer session), separate re-uploads of the same call, or some other cause — had not been confirmed against the original source recordings at the time of writing. Section 5 documents the interim policy adopted to combine these rows into a single company-quarter record, and Section 9 records this as an open item.

Data-quality note. Two further parsing issues were identified and are handled explicitly by the cleaning pipeline: (a) a small number of rows carry an implausible Year value produced by a filename-parsing edge case, and (b) nine rows in Dataset One carry a malformed 'Q#.error' quarter value on filenames that Dataset Two's more permissive parser reads correctly. Both are resolved during cleaning rather than left in the data as-is.

5. Data Cleaning & Integration Pipeline

Both raw datasets are combined once, by a standalone cleaning script, into a single canonical dataset. The dashboard reads only this canonical output — no cleaning logic runs inside the dashboard itself. The pipeline proceeds through seven steps, each auditable independently.

Step 1 — Load

Both raw CSV files are read, with company identifier columns loaded as nullable integers to avoid unintended type coercion.

Step 2 — Merge

Dataset Two (the superset, since it contains every transcript Dataset One has plus a handful of extraction failures) is left-joined with Dataset One on the Filename / File_Name key. A flag column, `has_climate_data`, is set based on whether a climate score exists for that row, immediately identifying the extraction-failure rows.

Step 3 — Year / Quarter Reconciliation

For every row, the parsed Year and Quarter from Dataset Two are used if they form a plausible period (a real quarter label together with a plausible year); if not, the pipeline falls back to Dataset One's parsed value under the same plausibility check. Only if neither source produces a plausible period is the row marked unparseable (`parse_ok = False`). Unparseable rows are never discarded — they are retained in the output but excluded from any period-based (year/quarter) view or filter in the dashboard, since their true period cannot be established.

Step 4 — Duplicate Resolution (interim policy)

Rows with a valid, reconciled period are grouped by company, year, and quarter. Where a group contains more than one contributing file, the raw countable metrics (sentence counts, word counts, category counts) are summed across the group. This function is deliberately isolated in the codebase as the single place to revisit once the meaning of the duplicate filenames is confirmed against source recordings; nothing else in the pipeline depends on this specific choice.

Step 5 — Recomputing Exact Ratios

After summing the raw counts for a company-quarter, every ratio-based metric is recalculated from those summed totals rather than averaged from the original per-file values, since these metrics are true ratios of countable quantities and can be reconstructed exactly: `Climate_Exposure_Score`, `Tone_Score`, `Percent_Uncertain`, `Percent_Weak_Modal`, and `File_Size_Readability`. This is applied uniformly to every row, including company-quarters with only one contributing file, so that one formula is used consistently across the entire dataset.

Step 6 — Approximating Combined Metrics

Two metrics are not exact ratios of countable quantities — FinBERT's probability outputs and the Fog Index — and cannot be exactly reconstructed after combining files. For company-quarters with multiple contributing files, these are instead combined as a word-count-weighted average across the group's files. This has the convenient side effect of automatically discounting any extraction-failure sub-file within a

group, since such a file contributes zero weight. Every row combined this way is flagged so this approximation is visible and traceable rather than silent.

Step 7 — Finalize

The cleaned, combined dataset is written out as the canonical output. The pipeline produced 15,329 final rows from 15,828 original source files (Dataset Two's full row count), of which 428 final rows represent a combined multi-file group, 20 rows are retained but flagged as period-unparseable, and 15 rows have no climate score available at all.

6. Canonical Clean Dataset Schema

The cleaned dataset contains 15,329 rows and 32 columns. This is the only dataset the dashboard reads.

Column	Description
Security_Code	BSE company identifier.
Company_Name / NSE_Name	Company name and short ticker-style name.
Year / Quarter	Reconciled fiscal period (see Step 3).
parse_ok	Whether a valid fiscal period could be established for this row.
n_source_files	Number of original transcript files combined into this row (1 unless a duplicate group).
Filenames	Semicolon-separated list of every source filename contributing to this row, for traceability.
duplicate_group	Whether this row combines more than one source file.
has_climate_data	Whether a climate score exists for this row (False only for extraction-failure files).
climate_file_coverage	Fraction of this row's contributing files that had a climate score (relevant only when duplicate_group is true).
Total_Sentences, Climate_Sentences_Count, Sentiment_Risk_Count, Sentiment_Opportunity_Count, Sentiment_Neutral_Count	Combined (summed) climate-scoring counts — see Dataset One.
Climate_Exposure_Score	Recomputed from the combined counts (Step 5).
Positive_Count, Negative_Count, Uncertain_Count, Weak_Modal_Count, Total_Words	Combined (summed) Loughran-McDonald counts and word count — see Dataset Two.
Tone_Score, Percent_Uncertain, Percent_Weak_Modal	Recomputed from the combined counts (Step 5).
File_Size_MB, File_Size_Readability	Combined file size and its recomputed readability proxy.
Fog_Index	Word-count-weighted average across contributing files (Step 6).
FinBERT_Positive, FinBERT_Negative, FinBERT_Neutral, FinBERT_Tone	Word-count-weighted average across contributing files (Step 6).

7. Column-to-Dashboard Mapping

This section maps every column in the canonical clean dataset to where it surfaces in the dashboard — as a filter, a chart, a table column, or supporting metadata. Dashboard page names refer to Section 8.

Column(s)	Where it appears in the dashboard
Company_Name, NSE_Name, Security_Code	Overview: distinct-company count. Company Explorer: the company search/selection control. Rankings: leaderboard row labels. Correlations: hover label on scatter points.
Year, Quarter	Overview: transcripts-per-year chart. Company Explorer: horizontal axis of every trend chart. Rankings & Correlations: year-range and quarter filter controls.
parse_ok	Governs which rows are eligible for any period-based chart or filter across all pages; unparseable rows for a selected company are shown separately as a footnote table on Company Explorer rather than silently omitted.
has_climate_data	Overview: ‘% of transcripts with a climate score’ metric tile. Company Explorer: triggers an explanatory notice in place of climate charts for a company with no climate-scored transcripts. Correlations: optional include/exclude filter.
n_source_files, Filenames, duplicate_group, climate_file_coverage	Company Explorer: shown in chart hover text and in the raw contributing-files table, so a combined multi-file quarter is always visibly distinguishable from a single-file one.
Climate_Exposure_Score	Overview: distribution histogram. Company Explorer: trend chart. Rankings: rankable metric. Correlations: correlation matrix and scatter axis option.
Tone_Score	Overview: distribution histogram. Company Explorer: trend chart. Rankings & Correlations: selectable metric.
FinBERT_Tone	Company Explorer: trend chart (plotted alongside Tone_Score for direct dictionary-vs-model comparison). Rankings & Correlations: selectable metric.
Fog_Index	Company Explorer: trend chart. Rankings & Correlations: selectable metric.
Percent_Uncertain, Percent_Weak_Modal	Rankings & Correlations: selectable metrics.
Total_Words	Rankings & Correlations: selectable metric (labeled Transcript Length).
Sentiment_Risk_Count, Sentiment_Opportunity_Count, Sentiment_Neutral_Count, Climate_Sentences_Count	Rankings & Correlations: selectable metrics.
Positive_Count, Negative_Count, Uncertain_Count, Weak_Modal_Count, Total_Sentences, File_Size_MB, File_Size_Readability, FinBERT_Positive, FinBERT_Negative, FinBERT_Neutral	Not exposed as a standalone chart metric; visible in the raw per-transcript data table on Company Explorer and documented in full on the About Data page for anyone auditing a specific figure.

8. Dashboard Architecture & Features

8.1 Technology Stack

The dashboard is a multi-page Streamlit application. Streamlit was chosen for direct compatibility with the pandas-based cleaning pipeline and for the fastest path from a tabular dataset to an interactive, filterable web interface. Charts are rendered with Plotly throughout, chosen specifically because two of the five pages (Correlations and Rankings) require chart types — an annotated correlation heatmap, a colour-coded scatter plot, and precisely labeled horizontal bar charts — that are not available through simpler built-in charting primitives. Using one charting library consistently across all five pages keeps the visual style uniform. The underlying cleaned dataset is loaded once and cached in memory, so filtering and navigating between pages is immediate.

8.2 Page: Overview

The landing page. Presents summary metric tiles (total companies covered, total transcripts processed, year range, share of transcripts with a valid fiscal period, share with an available climate score), a distribution histogram each for climate exposure and financial tone across the full dataset, and a bar chart of transcript volume per year. Serves as the orientation point before drilling into a specific company or comparison.

8.3 Page: Company Explorer

A search-and-select control lets the user pick any single company from the full universe covered by the data. Once selected, the page displays that company's climate exposure, financial tone, FinBERT tone, and readability scores plotted over time, one point per fiscal quarter. Hovering a point reveals how many source files were combined into it and whether it represents a duplicate-group approximation. A table beneath the charts lists every underlying transcript row contributing to that company's record, including any quarters excluded from the charts due to an unparseable fiscal period. If a selected company has no climate-scored transcripts at all, the page shows an explanatory notice in place of the climate charts rather than an empty or misleading plot.

8.4 Page: Rankings

Lets the user rank all companies by any one metric, over a chosen year range and set of quarters, as a Top-N or Bottom-N leaderboard. A minimum-data-volume filter prevents a company with only one or two matching transcripts from dominating an extreme ranking position. Results are shown as both a sortable table and a horizontal bar chart, chosen specifically because company names are long text strings that remain readable as horizontal labels.

8.5 Page: Correlations

Provides two complementary views for relating metrics to one another: a correlation matrix, rendered as an annotated heatmap, across a user-selected set of metrics; and a scatter-plot explorer where the user picks any two metrics for the horizontal and vertical axes, optionally coloured by year, at either the individual-transcript or company-averaged level. This page is where a question such as 'does higher climate exposure tend to coincide with more negative or more uncertain language' is investigated directly.

8.6 Page: About Data

A reference page documenting the full clean-schema column dictionary and the data-quality caveats recorded in Section 9 of this document, together with a brief description of how each dataset's scores were originally produced. Included so that any figure encountered elsewhere in the dashboard can be traced back to its definition and any relevant caveat without leaving the application.

9. Known Data Limitations & Open Items

- **Duplicate-file resolution is an interim policy.** The meaning of the numeric filename suffixes that create multi-file company-quarters has not yet been confirmed against source recordings. The current sum-and-recompute approach is a reasonable default but should be revisited once confirmed.
- **15 rows have no climate score.** These correspond to source files where text extraction failed upstream of both scoring pipelines; they carry zero counts across every climate and financial-tone metric and are flagged accordingly rather than silently imputed.
- **20 rows have an unrecoverable fiscal period** and are excluded from every year/quarter-based chart and filter, though retained in the dataset and visible in per-company detail views.
- **FinBERT and Fog Index values on combined multi-file rows are weighted-average approximations,** not exact recomputations, since neither metric is a simple ratio of countable quantities. This is flagged on every affected row via the `duplicate_group` indicator.
- **Dictionary-based scoring lacks contextual understanding.** The Loughran-McDonald approach counts words independent of surrounding context, and does not distinguish between different speakers on the same call (e.g. a CEO versus a CFO), nor compare stated language against actual reported financial results.

10. Appendix: Formula Reference

Metric	Formula
Climate_Exposure_Score	$\text{Climate_Sentences_Count} \div \text{Total_Sentences} \times 100$
Tone_Score	$(\text{Positive_Count} - \text{Negative_Count}) \div (\text{Positive_Count} + \text{Negative_Count})$
Percent_Uncertain	$\text{Uncertain_Count} \div \text{Total_Words} \times 100$
Percent_Weak_Modal	$\text{Weak_Modal_Count} \div \text{Total_Words} \times 100$
File_Size_Readability	$-\log(\text{File_Size_MB} + 1)$
Fog_Index	$0.4 \times (\text{average words per sentence} + \text{percent of words with 3+ syllables})$
FinBERT_Tone	$\text{FinBERT_Positive} - \text{FinBERT_Negative}$

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